House Price Prediction using Machine Learning

XYlofy AI Internship – Week 1 Project Summary

Submitted By: Shrishti Vaishnav

Project Objective

The objective of this project was to develop a machine learning model capable of predicting residential property prices using various housing characteristics such as area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and other amenities. The project aimed to demonstrate the complete machine learning workflow, including data exploration, preprocessing, model development, visualization, performance evaluation, and business insight generation for real estate price prediction.

Dataset Overview

The project utilized the **Housing Prices Dataset** obtained from Kaggle, containing **545 residential property records** and **13 original features**. The target variable is **price**, while the remaining attributes describe the structural and location-related characteristics of each house. Initial exploration confirmed that the dataset contained **no missing values and no duplicate records**, allowing for a straightforward preprocessing pipeline. Seven categorical variables were transformed into numerical representations using **One-Hot Encoding**, resulting in a final dataset consisting of **545 observations and 14 processed features** suitable for machine learning algorithms.

Model Performance

Two regression algorithms were implemented and compared:

• **Linear Regression**

• **Random Forest Regressor**

The Linear Regression model achieved the best overall performance with a **Mean Absolute Error (MAE) of $970,043.40**, a **Root Mean Squared Error (RMSE) of $1,324,506.96**, and an **R^2 score of 0.6529**. The Random Forest Regressor produced an **MAE of $1,021,546.04**, an **RMSE of $1,400,565.97**, and an **R^2 score of 0.6119**. Based on these evaluation metrics, Linear Regression demonstrated superior predictive accuracy and better generalization for this dataset.

Key Insights

Exploratory analysis and correlation visualization revealed that **property area** is the strongest factor influencing house prices, followed by the **number of bathrooms**, **number of stories**, and the presence of **air conditioning**. Features such as **preferred area** and **parking availability** also contribute positively to property valuation. Interestingly, amenities such as **guest rooms** and **basements** exhibit relatively weak relationships with price, indicating that buyers prioritize functional living space and essential infrastructure over additional specialty rooms. The price distribution is positively skewed, with most properties belonging to lower and middle price ranges while a small number of luxury houses create a long upper tail.

Business Recommendations

For real estate developers and investment firms seeking to maximize return on investment, resources should be allocated toward **expanding usable living space**, **adding or renovating bathrooms**, and **installing modern air conditioning systems**, as these improvements demonstrate the strongest positive impact on market value. The developed regression model can serve as an effective decision-support tool for automated property valuation, investment analysis, and preliminary pricing strategies, while final decisions should continue to incorporate professional market expertise and local real estate knowledge.

Conclusion

This project successfully implemented an end-to-end machine learning pipeline for house price prediction using Python, Pandas, Scikit-learn, Matplotlib, and Seaborn. Through comprehensive data preprocessing, visualization, and comparative model evaluation, the analysis identified the most influential property features and demonstrated that **Linear Regression provides the most accurate predictions with an R^2 score of 0.6529**, explaining approximately **65% of the variation in housing prices**. The findings illustrate how data-driven techniques can support informed real estate investment decisions and provide valuable insights into the factors that drive property valuation.